MARCO STEFANELLI

Meteorological and Oceanographic Data Scientist | Numerical Weather Prediction | Oceanographic modelling | Data Assimilation | Artificial Intelligence

- Nationality: Italian
- E-Mail: stfmrc.dev@gmail.com
- Personal webpage: https://stfmrc.github.io/MarcoStefanelli/
- Languages: Italian (native), English (fluent), Croatian (beginner)

PROFESSIONAL SUMMARY

Climate, Meteorological, and oceanographic data scientist with a PhD in "Earth System and Climate Science" and a Marie Skłodowska-Curie fellowship, focused on developing a neural-network-based 3DVar observation operator for weather radar assimilation to improve precipitation forecasts. Hands-on experience across climate modelling, numerical weather prediction (NWP), oceanographic modelling, model evaluation, data assimilation, precipitation forecast improvement, artificial intelligence, and Python-based scientific computing. Strong background in climate, atmospheric, and ocean modelling, with expertise in large-scale meteorological data analysis, Linux/HPC workflows, and the clear communication of scientific results to interdisciplinary teams.

CORE COMPETENCIES

Climate modelling | Oceanographic modelling | Numerical Weather Prediction (NWP) | Forecast validation and model evaluation | Severe precipitation and weather radar-based forecasting | Data assimilation | AI applications | Python (xarray, NumPy, PyTorch, TensorFlow, geopandas, rasterio) | Linux/Unix | HPC (SLURM, PBS, LSF) | Fortran | NetCDF/GRIB/HDF5 | Satellite and in situ observations.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher, SMASH (Marie Skłodowska-Curie COFUND), Faculty of Mathematics and Physics, University of Ljubljana | *Apr 2024 – Apr 2026*

- Designing a neural-network observation operator for weather radar data assimilation.
- Working with large meteorological observation datasets (reflectivity from weather radar), NWP model outputs, and forecast workflows in Python and Linux/HPC environments.
- Communicate research outcomes clearly to interdisciplinary teams.

Researcher, Euro-Mediterranean Centre on Climate Change (CMCC), Italy | *Feb 2023 – Mar 2024*

- Advanced ocean-model validation products against observations
- Developing ocean data assimilation methodologies for unstructured grid modelling.
- Working with large oceanographic observation datasets (ARGO and satellite data), unstructured model outputs, and operational workflows in Python and Linux/HPC environments.

Researcher, The Abdus Salam International Centre for Theoretical Physics (ICTP), Italy | *Apr 2018 – Apr 2019*

- Investigate the effects of urbanisation on the climate of the West African Monsoon.
- Applied atmospheric and climate modelling approaches to analyse land–atmosphere interactions.
- Working with large weather observation datasets (in situ precipitation and satellite data), climate model outputs, and operational workflows in Python and Linux/HPC environments.

VISITING

- **Imperial College London** - Visiting PhD student, Data Science Group, 3 months.
- **SISSA (Trieste)** - One-month collaboration on the SMASH postdoctoral project, developing a neural-network-based observation operator for weather radar assimilation.
- **ARSO (Ljubljana)** - One-month collaboration on the SMASH postdoctoral project, embedding the neural-network-based observation operator into a 3DVar data assimilation algorithm.

EDUCATION

- **PhD, Future Earth, Climate Change and Societal Challenges — University of Bologna & CMCC, Italy** | *Nov 2019 – Jan 2023*
  Dissertation: Data assimilation for advanced cross-scale unstructured-grid ocean modelling.
  Abstract: https://amsdottorato.unibo.it/id/eprint/10785/
  Download: https://amsdottorato.unibo.it/id/eprint/10785/1/Thesis_Stefanelli.pdf
  Related publication: https://doi.org/10.3389/fmars.2025.1656879

- **MSc, Earth System Physics — University of Trieste, Italy** | *Sep 2014 – Mar 2018*
  Dissertation: A study of urbanisation effects on the West-African monsoon.

- **BSc, Physics — University of Salento, Italy** | *Oct 2009 – Dec 2013*
  Dissertation: Mean sea level processes, observations, and evolution in the Mediterranean Sea.

SELECTED PUBLICATIONS

- Stefanelli, M., Jansen, E., Aydogdu, A., Federico, I., Coppini, G. (2025). Data Assimilation for Advanced Cross-Scale Unstructured-Grid Ocean Modelling. Frontiers in Marine Science.
  https://doi.org/10.3389/fmars.2025.1656879
- Stefanelli, M., Zaplotnik, Z., Skok, G. Preprint: A Neural Network-Based Observation Operator for Weather Radar Data Assimilation (under review).
  https://doi.org/10.48550/arXiv.2512.18289

SELECTED PRESENTATIONS

- Video (in italian) developed for the participation in the Premio Sergio Borghi at the Festival della Meteorologia in Rovereto
  Video: https://www.youtube.com/watch?v=R5q-pwXo9dY&t=7s
- European Geosciences Union (EGU), Wien (2022 and 2025)
  2022 - https://meetingorganizer.copernicus.org/EGU22/EGU22-4741.html
  2025 - https://meetingorganizer.copernicus.org/EGU25/EGU25-420.html
- International Conference on Alpine Meteorology (ICAM) (Poreč, Croatia) (2025)
- European Meteorological Society (EMS) Annual Meeting, Ljubljana (2025)
  Video: https://meetingorganizer.copernicus.org/EMS2025/EMS2025-28.html
- Slovenian Meteorology Society, Ljubljana (2025)
  Video: https://www.youtube.com/watch?v=kIYwkgOSsGA

ADDITIONAL INFORMATION

Grants: Marie Skłodowska-Curie COFUND postdoctoral grant (SMASH) | Puglia Region Youth Fund (PIN).

Tools and data: NetCDF libraries (NCO, CDO, NCAP), Jupyter, GitHub, GRIB/HDF5/NetCDF, weather radar, satellite, and in situ observations.

Educational outreach and science communication: Active in educational initiatives for high school students focused on climate change awareness, scientific literacy, and critical thinking for identifying misinformation. Developed a JavaScript interactive tool (free download on my website: **https://stfmrc.github.io/MarcoStefanelli/project/educational.html**) using the Lorenz-63 model to explore deterministic chaos and show how data assimilation can attenuate forecast uncertainty.